

RETAIL SALES AND PROFITABILITY ANALYSIS

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Tools Used: Excel, Pivot Tables, Pivot Charts, KPIs, Slicers



BUSINESS PROBLEM

The business was generating strong sales, but some products and regions were still causing profitability issues. The goal of this analysis was to identify loss-making areas and understand the factors reducing profit.”

DATASET OVERVIEW

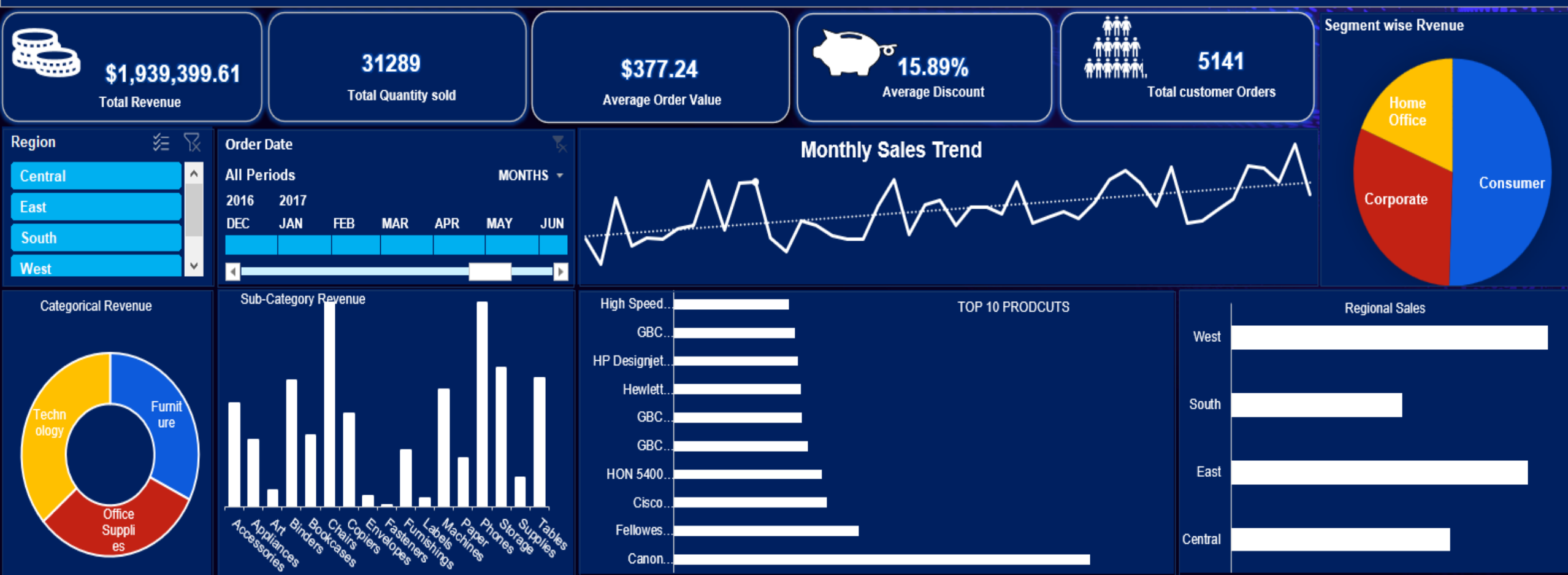
- ~9000 transaction records
- Sales, Profit, Region, Category, Quantity etc.
- Data validation performed using valid/invalid filter

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer	Customer Segment	Country	City	State	Postal Code	Region	Product	Category	Sub-Category	Product Name	Sales	Quantity	Discount	Profit	Shipping	Shipping Cost	Order Month	Order Year	Performance	
2718	CA-2014-100006	9/7/2014	9/13/2014	Standard Class	DK-13375	Dennis Kane	Consumer	United States	New York City	New York	10024	East	TEC-PH-10	Technology	Phonor	AT&T EL51110 DEC	377.97	3	0	109.6113	64	valid	Jul	2014	Good
6288	CA-2014-100090	7/8/2014	7/12/2014	Standard Class	EB-13705	Ed Braxton	Corporate	United States	San Francisco	California	94122	West	FUR-TA-10	Furniture	Tabler	Han 21111 Invitatio	502.488	3	0.2	-87.9354	122	valid	Aug	2014	Bad
6289	CA-2014-100090	7/8/2014	7/12/2014	Standard Class	EB-13705	Ed Braxton	Corporate	United States	San Francisco	California	94122	West	OFF-BI-10	Office Supplier	Binder	Wiken Jener Ledc	196.704	6	0.2	68.8464	122	valid	Aug	2014	Good
9515	CA-2014-100293	3/14/2014	3/18/2014	Standard Class	HF-18475	Neil Franzbrich	Home Office	United States	Jacksonville	Florida	32216	South	OFF-PA-10	Office Supplier	Paper	Nerax 1887	91.056	6	0.2	21.8696	4	valid	Mar	2014	Good
3824	CA-2014-100323	2/2/2014	2/3/2014	Standard Class	JC-15340	Jerry Casciopp	Consumer	United States	New York City	New York	10024	East	OFF-BI-10	Office Supplier	Binder	Praxburg Cass	3.923	0.2	0	1.3257	33	valid	Jan	2014	Good
3825	CA-2014-100323	4/8/2014	4/15/2014	Standard Class	JM-18455	Jim Mitchum	Corporate	United States	Glendale	Arizona	85201	West	OFF-FA-10	Office Supplier	Partonerr	Binder Cligby Ol	2.368	2	0.2	0.5233	-111	invalid	Aug	2014	Good
3827	CA-2014-100363	4/8/2014	4/15/2014	Standard Class	JM-15655	Jim Mitchum	Corporate	United States	Glendale	Arizona	85201	West	OFF-PA-10	Office Supplier	Paper	Thing In Da Tada	19.008	3	0.2	6.8904	-111	invalid	Aug	2014	Good
9441	CA-2014-100391	5/25/2014	5/29/2014	Standard Class	BW-11065	Barry Weirich	Consumer	United States	New York City	New York	10035	East	OFF-PA-10	Office Supplier	Paper	Strathmore Phat	14.62	2	0	6.7252	4	valid	May	2014	Good
6569	CA-2014-100678	4/18/2014	4/22/2014	Standard Class	KM-16720	Kunert Miller	Consumer	United States	Houston	Texas	77095	Central	OFF-AR-10	Office Supplier	Art	Prax Durtlazz CH	2.688	2	0.2	1.008	4	valid	Apr	2014	Good
6570	CA-2014-100678	4/18/2014	4/22/2014	Standard Class	KM-16720	Kunert Miller	Consumer	United States	Houston	Texas	77095	Central	FUR-OH-10	Furniture	Chair	DMi Artura Collec	317.058	3	0.3	-18.1176	4	valid	Apr	2014	Bad
6571	CA-2014-100678	4/18/2014	4/22/2014	Standard Class	KM-16720	Kunert Miller	Consumer	United States	Houston	Texas	77095	Central	OFF-EH-10	Office Supplier	Envelope	Cameo Buff Pallc	149.352	3	0.2	50.4062	4	valid	Apr	2014	Good
6572	CA-2014-100678	4/18/2014	4/22/2014	Standard Class	KM-16720	Kunert Miller	Consumer	United States	Houston	Texas	77095	Central	FUR-AC-10	Technology	Accessories	Crossing Exper	227.974	0.2	0	23.497	4	valid	Apr	2014	Good
9020	CA-2014-100706	12/16/2014	12/18/2014	Standard Class	LE-16310	Laurel Ellritan	Consumer	United States	Springfield	Virginia	22153	South	TEC-AC-10	Technology	Accessories	Care Lastic 2.4GH	99.98	2	0	7.9934	2	valid	Dec	2014	Good
9021	CA-2014-100706	12/16/2014	12/18/2014	Standard Class	LE-16310	Laurel Ellritan	Consumer	United States	Springfield	Virginia	22153	South	FUR-FU-10	Furniture	Furnishing	Ultra Daser Push P	29.46	6	0	9.7218	2	valid	Dec	2014	Good
6315	CA-2014-100762	11/24/2014	11/29/2014	Standard Class	NG-19355	Nat Gilpin	Corporate	United States	Jackson	Michigan	49201	Central	OFF-AR-10	Office Supplier	Art	Hunt Pauer Houz	151.92	4	0	45.576	5	valid	Nov	2014	Good
6316	CA-2014-100762	11/24/2014	11/29/2014	Standard Class	NG-19355	Nat Gilpin	Corporate	United States	Jackson	Michigan	49201	Central	OFF-LA-10	Office Supplier	Label	Hut Matrix Printe	196.62	2	0	96.2438	5	valid	Nov	2014	Good
6317	CA-2014-100762	11/24/2014	11/29/2014	Standard Class	NG-19355	Nat Gilpin	Corporate	United States	Jackson	Michigan	49201	Central	OFF-PA-10	Office Supplier	Paper	Nerax 1885	144.12	3	0	69.1776	5	valid	Nov	2014	Good
6318	CA-2014-100762	11/24/2014	11/29/2014	Standard Class	NG-19355	Nat Gilpin	Corporate	United States	Jackson	Michigan	49201	Central	OFF-PA-10	Office Supplier	Paper	Adam Telephone	15.96	2	0	7.93	5	valid	Nov	2014	Good
9440	CA-2014-100828	3/23/2014	3/28/2014	Standard Class	OS-12505	Grady Stewart	Consumer	United States	Pomona	California	91767	West	OFF-LA-10	Office Supplier	Label	Emad Alkash 2.0	18.75	5	0	9	4	valid	Mar	2014	Good
9462	CA-2014-100887	10/19/2014	10/24/2014	Standard Class	EH-14125	Eugene Hildebr	Home Office	United States	Los Angeles	California	90712	West	TEC-PH-10	Technology	Phonor	ATA Wipz Integre	321.552	6	0.2	20.097	5	valid	Oct	2014	Good
8464	CA-2014-100881	3/23/2014	4/1/2014	Standard Class	DR-12940	Daniel Raglin	Consumer	United States	Albuquerque	New Mexico	87105	West	FUR-FU-10	Technology	Phonor	AT&T TR1909W	302.376	3	0.2	22.6782	-83	invalid	Mar	2014	Good
3888	CA-2014-100895	6/2/2014	6/6/2014	Standard Class	SV-20735	Stewart Virinbk	Consumer	United States	Ranuell	Georgia	30076	South	OFF-AR-10	Office Supplier	Art	Sanford Califoric	8.56	2	0	2.6536	120	valid	Feb	2014	Good
3889	CA-2014-100895	6/2/2014	6/6/2014	Standard Class	SV-20735	Stewart Virinbk	Consumer	United States	Ranuell	Georgia	30076	South	TEC-PH-10	Technology	Phonor	Maghie Juice Pac	239.97	3	0	67.1916	120	valid	Feb	2014	Good
3890	CA-2014-100895	6/2/2014	6/6/2014	Standard Class	SV-20735	Stewart Virinbk	Consumer	United States	Ranuell	Georgia	30076	South	OFF-ST-10	Office Supplier	Storage	Hut File 7-Packe	356.94	2	0	107.052	120	valid	Feb	2014	Good
7572	CA-2014-100916	10/21/2014	10/26/2014	Standard Class	FH-14275	Frank Hausley	Corporate	United States	Newport	Virginia	23602	South	OFF-ST-10	Office Supplier	Storage	Eldan Bare farate	194.7	5	0	9.735	5	valid	Oct	2014	Good
7573	CA-2014-100916	10/21/2014	10/26/2014	Standard Class	FH-14275	Frank Hausley	Corporate	United States	Newport	Virginia	23602	South	OFF-ST-10	Office Supplier	Storage	Han 21111 Invitatio	591.32	4	0.4	112.368	5	valid	Oct	2014	Good
7574	CA-2014-100916	10/21/2014	10/26/2014	Standard Class	FH-14275	Frank Hausley	Corporate	United States	Newport	Virginia	23602	South	OFF-AR-10	Office Supplier	Art	Nerax 1885	144.12	3	0	69.1776	5	valid	Oct	2014	Good
7868	CA-2014-100972	11/19/2014	11/24/2014	Standard Class	DE-13340	Dennis Balton	Home Office	United States	Salt Lake City	Utah	84106	West	OFF-PA-10	Office Supplier	Paper	Nerax 1885	166.44	3	0	79.9912	5	valid	Nov	2014	Good
6252	CA-2014-101147	12/2/2014	12/4/2014	First Class	MC-17575	Matt Collins	Consumer	United States	Chicago	Illinois	60623	Central	OFF-AP-10	Office Supplier	Appliances	Staple holder	2.394	1	0.8	-6.3441	59	valid	Feb	2014	Bad
4565	CA-2014-101175	12/9/2014	12/14/2014	Standard Class	DM-12955	Daria Medina	Corporate	United States	Mesa	Arizona	85204	West	OFF-ST-10	Office Supplier	Storage	Acce Perma 3000	100.704	6	0.2	-1.2588	93	valid	Sep	2014	Bad
6170	CA-2014-101266	8/27/2014	8/30/2014	Standard Class	MM-17920	Michael Moore	Consumer	United States	Springfield	Virginia	22153	South	OFF-PA-10	Office Supplier	Paper	Nerax 1898	13.26	2	0	6.4128	3	valid	Aug	2014	Good
9724	CA-2014-101364	12/2/2014	12/2/2014	Standard Class	TW-21025	Tamara Willingh	Home Office	United States	New York City	New York	10011	East	OFF-BI-10	Office Supplier	Binder	Lock-Up Eazol'Sp	296.742	13	0.2	100.1403	4	valid	Dec	2014	Good
3941	CA-2014-101392	12/7/2014	12/13/2014	Standard Class	AS-10630	Ann Steele	Home Office	United States	Seattle	Washington	98103	West	OFF-ST-10	Office Supplier	Storage	Adventur Rellina	249.36	7	0	78.4326	154	valid	Jul	2014	Good
6190	CA-2014-101427	12/2/2014	12/2/2014	Standard Class	AY-10595	Andy Yatev	Corporate	United States	Los Angeles	California	91149	East	OFF-AR-10	Office Supplier	Art	Eldan Obatliqui	1.016	3	0.3	1.1022	4	valid	Dec	2014	Good
6850	CA-2014-101462	4/20/2014	4/20/2014	Standard Class	BP-11230	Benjamin Patte	Consumer	United States	Lar Angeles	California	90045	West	FUR-FU-10	Furniture	Furnishing	GE 4 Fast Flowe	59.92	4	0	27.5632	5	valid	Apr	2014	Good
428	CA-2014-101476	9/12/2014	9/12/2014	First Class	SD-20485	Shirley Daniel	Home Office	United States	New Rochelle	New York	10801	East	TEC-MA-10	Technology	Machine	Egran Work Force	69.99	1	0	30.0957	-87	invalid	Dec	2014	Good
5193	CA-2014-101560	11/2/2014	11/2/2014	Standard Class	CS-12250	Chris Solernick	Corporate	United States	Columbia	South Carolina	29203	South	FUR-FU-10	Furniture	Furnishing	Eldan Cleatmat P	397.6	5	0	43.736	-320	invalid	Nov	2014	Good
5194	CA-2014-101560	11/2/2014	11/2/2014	Standard Class	CS-12250	Chris Solernick	Corporate	United States	Columbia	South Carolina	29203	South	OFF-PA-10	Office Supplier	Paper	Nerax 1923	85.96	7	0	40.4012	-320	invalid	Nov	2014	Good
5195	CA-2014-101560	11/2/2014	11/2/2014	Standard Class	CS-12250	Chris Solernick	Corporate	United States	Columbia	South Carolina	29203	South	OFF-AR-10	Office Supplier	Art	Crayola Callers dF	13.12	4	0	4.3266	-320	invalid	Nov	2014	Good
5196	CA-2014-101560	11/2/2014	11/2/2014	Standard Class	CS-12250	Chris Solernick	Corporate	United States	Columbia	South Carolina	29203	South	OFF-BI-10	Office Supplier	Binder	GEBC Linn Leap W	45.65	3	0	22.2374	-320	invalid	Nov	2014	Good
1571	CA-2014-101624	12/15/2014	12/15/2014	First Class	MC-18100	Mike Orbach	Consumer	United States	Chicago	Illinois	79907	Central	OFF-AR-10	Office Supplier	Art	LifeTime Advant	763.28	5	0.3	-21.808	3	valid	Dec	2014	Bad
1574	CA-2014-101602	12/15/2014	12/15/2014	First Class	MC-18100	Mike Orbach	Consumer	United States	El Paso	Texas	79907	Central	FUR-OH-10	Furniture	Chair	LifeTime Advant	763.28	5	0.3	-21.808	3	valid	Dec	2014	Bad
5480	CA-2014-101770	3/31/2014	4/4/2014	Standard Class	KB-16240	Karen Bern	Corporate	United States	Miami	Florida	33180	South	OFF-BI-10	Office Supplier	Binder	Avery Hale Reinfr	1.869	1	0.7	-1.3083	4	valid	Mar	2014	Bad
9461	CA-2014-101833	11/17/2014	11/22/2014	Standard Class	FG-14260	Frank Gattineau	Home Office	United States	New York City	New York	10024	East	OFF-PA-10	Office Supplier	Paper	Southworth 100%	34.44	3	0	17.22	5	valid	Nov	2014	Good
3843	CA-2014-101931	10/28/2014	10/31/2014	First Class	TS-21370	Tedd Sumrell	Corporate	United States	Lar Angeles	California	90049	West	OFF-BI-10	Office Supplier	Binder	Avery Nan-Stick	7.184	2	0.2	2.245	3	valid	Oct	2014	Good
3844	CA-2014-101931	10/28/2014	10/31/2014	First Class	TS-21370	Tedd Sumrell	Corporate	United States	Lar Angeles	California	90049	West	OFF-SU-10	Office Supplier	Supplier	Serrated Blade at	6.28	2	0	0.0628	3	valid	Oct	2014	Good
3845	CA-2014-101931	10/28/2014	10/31/2014	First Class	TS-21370	Tedd Sumrell	Corporate	United States	Lar Angeles	California	90049	West	OFF-SU-10	Office Supplier	Supplier	Premier Automat	480.74	2	0	14.4222	3	valid	Oct	2014	Good
3846	CA-2014-101931	10/28/2014	10/31/2014	First Class	TS-21370	Tedd Sumrell	Corporate	United States	Lar Angeles	California	90049	West	OFF-PA-10	Office Supplier	Paper	O'Sullivan Living	616.446	3	0.15	-3.9632	3	valid	Oct	2014	Good
3847	CA-2014-101931	10/28/2014	10/31/2014	First Class	TS-21370	Tedd Sumrell	Corporate	United States	Lar Angeles	California	90049	West	OFF-ST-10	Office Supplier	Storage	Eldan Obatliqui	1.016	3	0.3	1.1022	3	valid	Oct	2014	Good
1791	CA-2014-102008	9/20/2014	10/4/2014	Standard Class	RA-19915	Russell Appleac	Consumer	United States	New York City	New York	10011	East	OFF-PA-10	Office Supplier	Paper	Tapr Green Bar G	48.94	1	0	24.47	-173	invalid	Sep	2014	Good
4948	CA-2014-102085	10/4/2014	10/9/2014	Standard Class	JD-16015	Jay Daniels	Consumer	United States	Seattle	Washington	98115	West	OFF-LA-10	Office Supplier	Label	Avery 519	29.24	4	0	13.7428	153	valid	Apr	2014	Good
3399	CA-2014-102274	11/21/2014	11/26/2014	Standard Class	DM-13075	Dave Halliten	Corporate	United States	Richmond	Kentucky	40475	South	TEC-PH-10	Technology	Phonor	Legitech B530 US	36.99	1	0	9.9873	5	valid	Nov	2014	Good

Sales Analysis

- West region generated highest sales
- Customer segment contributed most revenue
- Phones were top revenue –driving product

RETAIL SALES REVENUE DASHBAORD



PROFITABILITY ANALYSIS

\$249,505.81

Total Profit

13%

Profit Margin

\$48.53

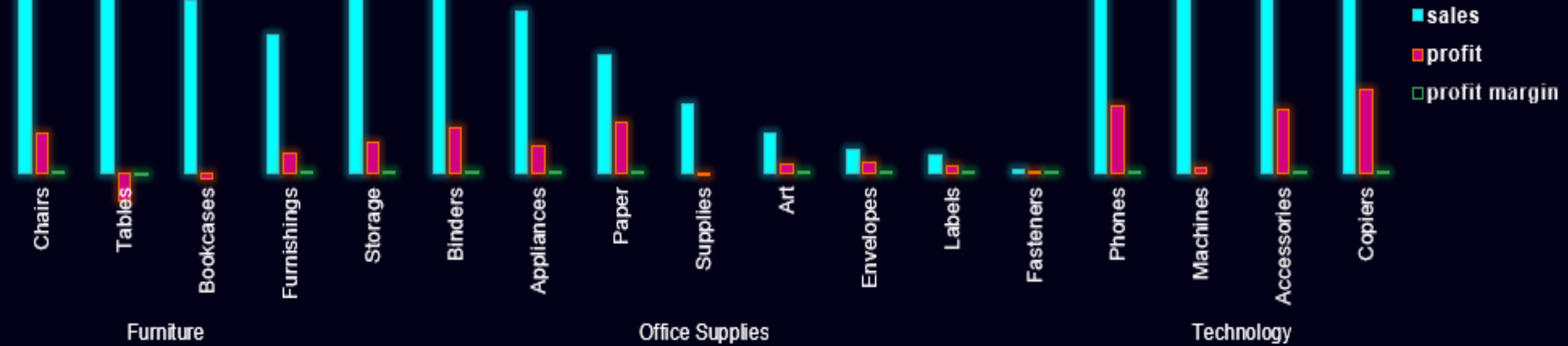
Avg Order Profit

16%

Loss Making Orders

1 out of 6 orders is loss making

sub-category wise sales and profit comparison



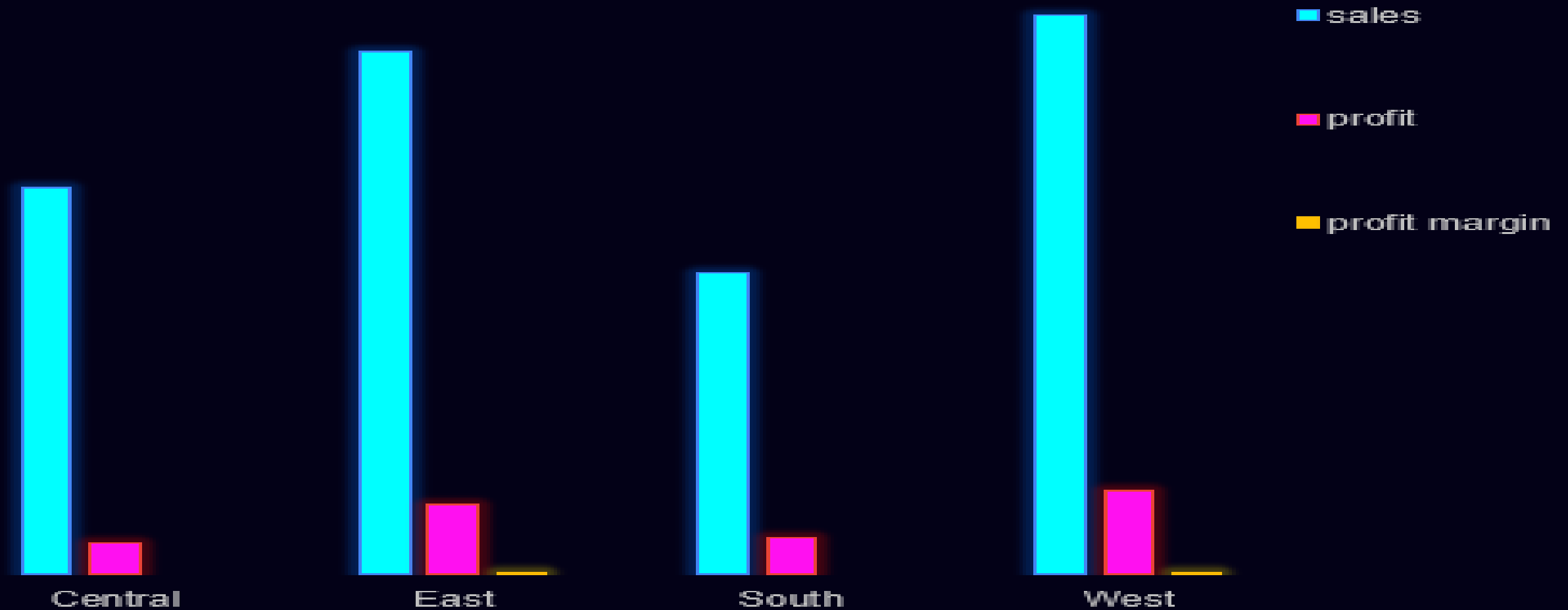
Phone is profit driving sub-category

Chairs are low margin product

Table is biggest concern of the business ,because it generates highest lost with decent revenue contribution

Papers, Accessories and copiers are potential growth sub-category

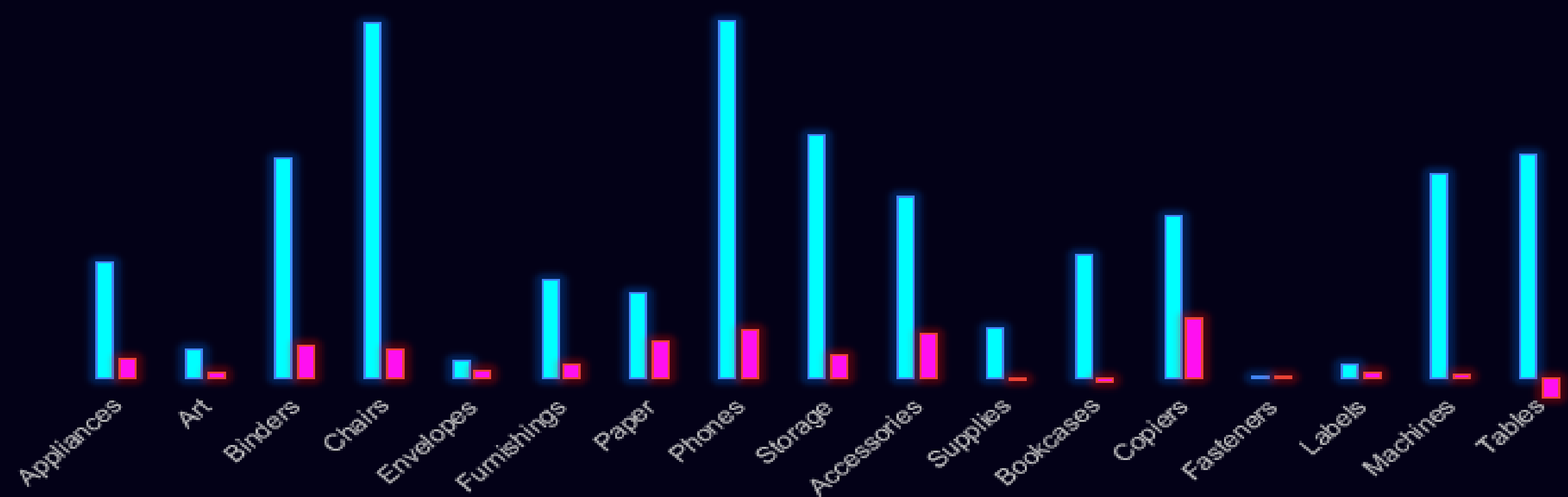
Region wise profitability analysis



Central region appears inefficient because despite stronger sales contribution than South, it generates weaker profitability, making it a potential **profitability-risk** region for further investigation.

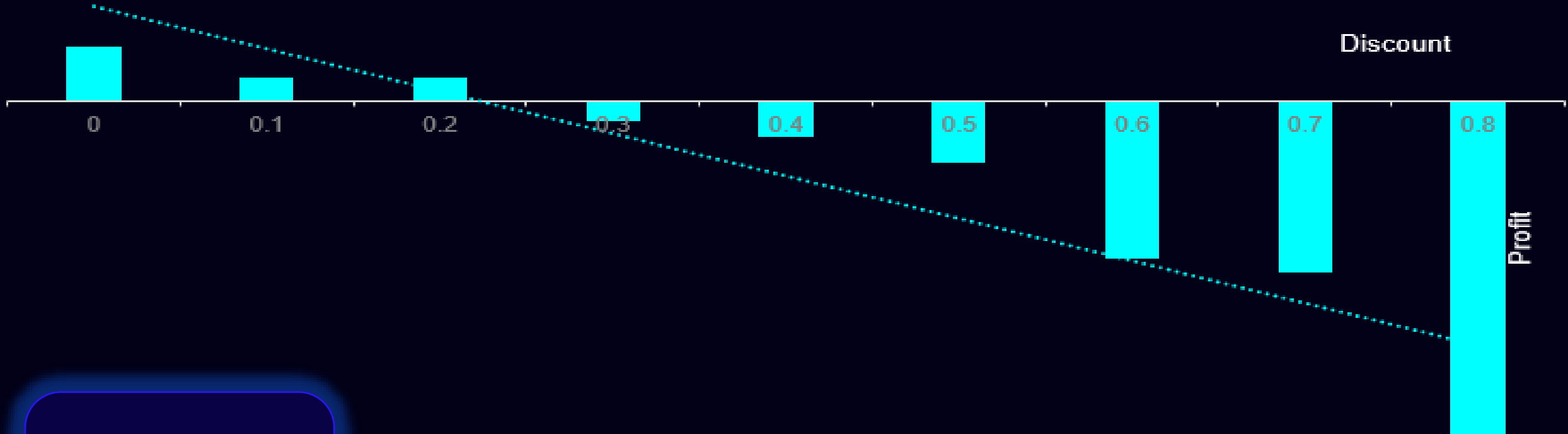
Root CAUSE ANALYSIS

Sub-Category Performance in central region



Tables and Bookcases are eroding profit

Profit v/s Discount Behaviour



15%

AVG DISCOUNT

“Excessive discounting beyond sustainable levels is driving those losses.”

Top Loss-Making Cities (Central Region)



KEY INSIGHTS

- Around 16% of orders are making losses, showing that not all sales are helping the business earn profit.
- Phones, Copiers, and Paper are strong-performing sub-categories because they generate both good sales and good profit.
- Tables and Bookcases are major problem areas because they generate decent sales but still result in losses.
- Central region is less efficient because it generates higher sales than South region but lower profit.
- In the Central region, sub-categories like Tables, Appliances, Furnishings, and Bookcases are mainly responsible for losses and also have higher average discounts.
- Discounts around 20% still help generate healthy sales and profit, but profits start falling after 30% discounts.
- Very high discounts between 50%–80% create heavy losses without generating enough sales benefit, making them risky for the business.

Recommendations

- Reduce heavy discounts on products like Tables and Bookcases because these products are bringing sales but also causing major losses.
- Try to keep discounts below 30% whenever possible, since profits start dropping sharply after that point.
- Avoid giving extremely high discounts (50%–80%) because they create big losses without bringing enough extra sales.
- Review the Central region more carefully, since it is generating good sales but weaker profits compared to other regions.
- Focus more on products like Phones, Copiers, and Paper because they are generating both good sales and healthy profits.
- Regularly track loss-making orders and profit margins so business problems can be identified early before losses increase.

“The project demonstrates
how data analysis can support
profitability optimization and
business decision-making”

THANK
YOU